Retail Customer Behavior for Growth

Phase 2: Polyglot Persistence & Predictive Analytics

Real World Data: Dunnhumby Complete Journey Dataset

MongoDB Customer 360 + Predictive Modelling

PostgreSQL Schema & Ingestion

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Business Case — Why This Analysis Matters

The Opportunity

Competitive Advantage

Retailers who address different spending mindsets can differentiate themselves by providing a shopping experience that resonates with consumers on a deeper level.

Market Adaptability

Awareness of shifts in consumer spending patterns helps retailers adapt to changing market conditions — including inflationary and tariff pressures.

Revenue Optimization

Understanding customer dynamics and spending patterns enables retailers to optimize marketing strategies and improve sales performance.

Stakeholder Questions

- How much and when are customers spending?
- Which demographic segments drive the most value?
- Which product categories are growing fastest?
- What is the true incremental impact of campaigns?

Dataset: Dunnhumby Complete Journey

Transactions	~2.5M rows · household-level purchase history
Campaigns	30 campaigns · 4,883 household-campaign records
Demographics	924 households · 7 anonymized classifications
Coupons	119,384 coupon pool · 2,318 redemptions
Products	Dept · Commodity · Sub-commodity hierarchy
Causal data	Weekly mailer & in-store display flags

Research Questions — Phase 1 Commitments & Phase 2 Outcomes

First-Order Questions (committed in Phase 1)

Track customer spending trends

PostgreSQL ✓

Wks 1–16 ramp artefact identified; steady state $75K–$95K/wk; Wk 102 truncation noted

Evaluate demographic influences

PostgreSQL ✓

classification_3 strongest signal (~$7K spread); classification_2 Group X +$1K premium

Identify growth categories

PostgreSQL ✓

GROCERY ~60–70% of revenue; top 5 depts = 85%+ of sales; DRUG GM #2

Measure campaign impact

MongoDB + ML ✓

Type C +$449/+141% raw; OLS R^2=0.094 after controls; selection bias quantified (14× gap)

Second-Order Questions (Phase 2 extensions — Burns G.)

- ✓ Does macro-economic pressure (inflation/PPD) predict churn? → Weather enrichment raises PR-AUC from 0.16 → 0.63 (Model B confirmed significant)
- ✓ Do high-inflation campaign windows reduce uplift? → avg_ppd_during_campaign coef = −17.59 (p=0.010) — confirmed significant negative effect
- ~ Is there measurable trade-down to budget-tier products? → PPD main effect coef = +0.0027 (p=0.286 — directional but not statistically significant)

Second-Order Questions — Phase 2 Extensions

Three exploratory hypotheses generated from Phase 1 findings — tested with external data enrichment and ML

1 Does macro-economic pressure (inflation / PPD) predict churn?

✓ Confirmed significant

Hypothesis

Households experiencing sustained purchasing power deterioration would reduce trip frequency, creating a behavioural churn signal distinct from voluntary disengagement.

Method

XGBoost churn model (binary: 30-day absence proxy). Model A: internal features only. Model B: + weather (rain_days_last90d, heat_days_last90d, avg_temp) + PPD index.

Python · XGBoost · SHAP · Open-Meteo API · FRED CPI

Result

CONFIRMED — weather enrichment raises PR-AUC from 0.16 → 0.63 (4× improvement). rain_days_last90d (SHAP 1.58) and heat_days_last90d (SHAP 1.57) are the top 2 features — 4× more influential than any behavioural feature.

Households who shopped through bad weather are habitual buyers — rain exposure is a strong negative churn predictor (protective signal). Extreme heat raises churn risk non-linearly — heatwave-driven absences.

PR-AUC is the correct metric here (only 4.6% churners). ROC-AUC improvement (0.83→0.91) undersells the gain — PR-AUC improvement from near-chance to genuinely useful.

2 Do high-inflation campaign windows reduce campaign uplift?

✓ Confirmed significant (p=0.010)

Hypothesis

Campaigns running during periods of elevated inflation (high PPD) would be less effective as household purchasing power is constrained — inflation competes directly with promotional stimulus.

Method

OLS regression on DiD spend lift per campaign event. Controls: avg_ppd_during_campaign, avg_temp_campaign, cold_days_campaign, campaign_days, campaign_type. 4,165 campaign events.

Python · statsmodels OLS · ext_macro_weekly PPD index

Result

CONFIRMED — avg_ppd_during_campaign coef = −17.59 (p=0.010). Every 1pp rise in the PPD index during a campaign window reduces expected household spend lift by ~$18. Temperature and cold days also significant.

Campaign TypeB vs TypeA: coef −17.54 (p=0.069, n.s.) — the format gap disappears once inflation is controlled. Campaign TypeC vs TypeA: coef −4.36 (p=0.728, n.s.) — TypeC advantage is inflation-environment driven, not format-driven.

R^2=0.094 — acceptable given many spend drivers outside the model. The inflation coefficient is the key finding: it validates the PPD enrichment and reframes the campaign hierarchy result.

3 Is there measurable trade-down to budget-tier products?

↗ Directional — not significant

Hypothesis

Sustained inflationary pressure (rising PPD) would cause households to shift budget share toward lower-priced commodity tiers — a classic consumer trade-down behaviour.

Method

258 commodities assigned to 4 price quartile tiers (Budget $1.35 / Low-Mid $2.33 / High-Mid $3.22 / Premium $6.65). OLS regression: budget_share ~ PPD × commodity_tier × household fixed effects.

Python · statsmodels OLS · PostgreSQL commodity price aggregations

Result

NOT CONFIRMED (directional only). PPD main coef = +0.0027 (p=0.286) — positive direction as hypothesised but not statistically significant. High vs Mid-tier structural gap coef = −0.034 (p=0.030) — significant but pre-existing, not PPD-driven. PPD×Group interaction: p>0.20 (n.s.).

The absence of significance is a finding in itself: this dataset (2 years, 801 households, 5.5% PPD range) may lack the statistical power to detect trade-down. A longer window or sharper inflation shock (e.g., post-Liberation Day) might reveal the effect.

Anchor examples by tier: Budget = store-brand canned goods ($1.35 avg); Premium = branded organic produce, premium deli ($6.65 avg). The structural High vs Mid gap (p=0.030) confirms Households systematically under-allocate to the highest tier — independent of PPD.

Phase 2 — Polyglot Persistence Architecture

PostgreSQL

Transactional core

Fact tables · Marts

External ref tables

MongoDB

Customer 360

Household documents

ML scores · Geo

External Data Enrichment (Burns — Phase 2 Modelling)

- ext_calendar (711 rows) · Day-integer → calendar date mapping for all cross-system joins
- ext_macro_weekly (~102 rows) · FRED CPI: PPD index, food, core, apparel inflation (Jan 2023 – Dec 2024)
- ext_weather_daily (~711 rows) · Open-Meteo Allen TX: temp, precipitation, extreme-heat/cold flags

PostgreSQL — Source of Truth for MongoDB Hydration & External Enrichment

Burns (G.) · PostgreSQL acted as the single source of truth: powering all five MongoDB hydration passes and hosting the shared macro & weather reference tables.

Used for hydrating Mongodb and storing external data

- Star schema — fact_transactions, dim_households, dim_products, dim_departments, dim_stores, dim_date
- ERD highlighting key FKs and cardinality (~2.5M transactions · 924 households · 92K products)
- Ingestion pipeline — CSV loading, constraint validation, upsert strategy, row counts per table
- Mart tables — mart_churn_scores, mart_household_segments, mart_household_metrics
- External reference tables — ext_calendar, ext_macro_weekly, ext_weather_daily (shared reusable layer)
- Indexing strategy and query performance benchmarks

PostgreSQL is the source-of-truth for all four databases. ext_macro_weekly and ext_weather_daily are shared reference tables.

PostgreSQL — Star Schema Architecture

fact_transactions at the centre · dimension tables radiating out · mart & external tables for analytics

dim_households
- 🔑 household_key
- age_range
- marital_status
- income_range
- homeownership
- household_comp
- household_size
- kid_category

dim_date
- 🔑 day_key
- calendar_date
- week_no
- month · year · season
- is_weekend · is_holiday

dim_stores
- 🔑 store_id
- store_name
- location_type
- lon · lat (GeoJSON)

dim_demographics
- 🔑 household_key
- classification_1
- classification_2
- classification_3 ← key signal
- classification_4
- classification_5
- classification_6
- classification_7

fact_transactions
- 🔑 transaction_id
- 🔗 household_key
- 🔗 product_id
- 🔗 store_id
- 🔗 day_key → dim_date
- 🔗 campaign_id
- basket_id
- quantity
- sales_value
- retail_disc
- coupon_disc
- coupon_match_disc

dim_campaigns
- 🔑 campaign_id
- campaign_type (A/B/C)
- start_day · end_day
- duration_days
- description

coupon / coupon_redempt
- 🔑 coupon_upc
- 🔗 campaign_id
- 🔗 product_id
- redemption_date
- 119,384 coupons

dim_products
- 🔑 product_id
- 🔗 department_id
- 🔗 commodity_id
- 🔗 sub_commodity_id
- manufacturer
- brand
- curr_size_of_product

dim_departments / commodities / sub_commodities
- 🔑 dept/commodity/sub_commodity_id
- department_name · commodity_desc · sub_commodity_desc

Mart & Score Tables
- mart_household_metrics
- mart_churn_scores
- mart_household_segments
- ———
- ext_calendar (711r)
- ext_macro_weekly
- ext_weather_daily

🔑 Primary Key 🔗 Foreign Key Teal = core transaction dims Purple = demographics Amber = product hierarchy Blue = store Red = campaigns Mint = mart/external

MongoDB Customer 360 — Database Design

Document Schema — retail_customer360.households

Identifiers	
`_id`	HH_{household_key}
`household_key`	Integer PK (matches PostgreSQL)
Demographics	
`demographics{}`	classification_1 … classification_7
Financial Metrics	
`financial_metrics{}`	lifetime_spend · avg_basket_value
	total_trips · avg_inter_purchase_days
	days_since_last_purchase · coupon_rate
RFM	
`rfm_scores{}`	R, F, M quintiles + rfm_code
`rfm_segment`	Champions/Loyal/At Risk/…
ML Scores	
`ml_scores{}`	churn_risk_score · churn_label
	clv_segment · model_version · scored_at
Exogenous Context	
`exog_context{}`	ppd_at_last_purchase · food_inflation
	avg_temp_last90d · rain_days · heat_days
Arrays	

Indexes

2dsphere	`idx_store_2dsphere` $near · $geoNear · $geoWithin
Compound	`idx_churn_clv` Campaign targeting (churn_label + clv_segment)
Single	`idx_segments` Segment membership lookup
Single	`idx_recency` Overdue/churn cohort queries
Single	`idx_lifetime_spend` High-value leaderboard queries
Single	`idx_rfm_segment` RFM fan-out / targeting queries

Query Latency Benchmarks (50 runs · local Docker)

Query	Avg ms	Median	P95
Single HH lookup (find_one)	1.25	1.15	1.57
Recency count_documents	1.85	–	2.66
Recency find+sort+limit(10)	3.69	–	5.76
Segment aggregation ($unwind)	4.67	–	6.58
Segment count (indexed)	1.36	–	1.81
Spend leaderboard top-10	7.20	–	12.66

MongoDB Customer 360 — Document Structure

One document per household · All context co-located · No joins required at query time

households { ... } — one document per household

_id:	"HH_{household_key}"
household_key:	integer
demographics:	{ ... } →
classification_1..7:	anonymised attrs
financial_metrics:	{ ... } →
lifetime_spend:	$0 – $15K
avg_basket_value:	per visit
total_trips:	trip count
coupon_redemption_rate:	0–1
rfm_segment:	"Champions" | "At Risk" | ...
rfm_scores:	{ R, F, M, rfm_code }
ml_scores:	{ ... } →
churn_risk_score:	0.0 – 1.0
churn_label:	"Active" | "At Risk"
clv_segment:	"High" | "Mid" | "Low"
model_version:	"xgb_v3_clean_weather"
exog_context:	{ ... } →
ppd_at_last_purchase:	0.0–0.06
avg_temp_last90d:	°C
rain_days_last90d:	count
nearest_store:	{ store_id, location: GeoJSON Point }
preferred_categories:	[array]
recent_transactions:	[last 10]
active_campaigns:	[array]
created_at / updated_at / etl_version:	metadata

6 Indexes

idx_store_2dsphere — 2dsphere
Field: nearest_store.location
Use: Geo proximity queries ($geoNear)

idx_churn_clv — compound
Field: ml_scores.churn_risk + clv
Use: Retention targeting — CLV × churn

idx_rfm_segment — single
Field: rfm_segment
Use: Segment fan-out queries

idx_recency — single
Field: financial_metrics.days_since...
Use: Lapsed / Win-Back audience filter

idx_lifetime_spend — single
Field: financial_metrics.lifetime_spend
Use: High-value leaderboard queries

idx_segments — single
Field: segments array
Use: Multi-segment audience builder

5-Pass Hydration Pipeline

PostgreSQL (SSoT)

fact_transactions
2.5M rows

dim_households
dim_demographics
dim_stores
dim_campaigns

mart tables
ext_calendar
ext_macro
ext_weather

1. **Base Hydration** — 801 docs · 138/s
   Demographics + GeoJSON nearest store + ETL lineage
2. **Financial Metrics** — ~80 min · bottleneck
   Live aggregation over 2.5M rows · chunked batch 200
3. **Campaign Enrichment** — 192s · 4.6 HH/s
   active_campaigns array per household · 883/924 matched
4. **RFM Scoring** — Instant
   Quintile scoring from financial_metrics · no DB round-trip
5. **ML Write-back** — 0.5s · 1,602/s
   churn_risk_score + clv_segment + exog_context embedded

Demographics · ML Scores · RFM · Financial Metrics · Geo/Spatial · Exogenous Context · Array fields

MongoDB Customer 360 — Hydration & Enrichment Pipeline

Dunnhumby CSVs

households · transactions

products · stores

Campaign CSVs

campaign_table · coupon

coupon_redempt

FRED / BLS

CPI-U · Food CPI

Core · Apparel CPI

Open-Meteo API

Allen TX 33.1°N

temp · precip · flags

XGBoost & K-Means Models

churn scores · CLV

SHAP features

PostgreSQL — retail_analytics

fact_transactions · dim_households · dim_products · dim_stores
campaign_table · coupon_redempt · dim_campaigns

PostgreSQL — External Reference Tables

ext_calendar (711 rows) · ext_macro_weekly (~102 rows) · ext_weather_daily (~711 rows)

mongo_ingest.py

Pass 1

Base Hydration

demographics · nearest_store

GeoJSON · etl_version

Pass 2

Financial Metrics

lifetime_spend · avg_basket

total_trips · recency · coupon

Pass 3

Campaign Enrichment

active_campaigns array

coverage: 883/924 (95.6%)

Pass 4

RFM Scoring

R/F/M quintiles · rfm_code

rfm_segment label

Pass 5 — ML Write-back (BurnsG_modelling.ipynb)

- ml_scores.churn_risk_score + churn_label (xgb_v3_clean_weather)
- ml_scores.clv_segment (kmeans_v1)
- exog_context: ppd_at_last_purchase · food_inflation · avg_temp_last90d · rain_days · heat_days

External Data Enrichments — Why They Are Non-Negotiable

Three shared reference tables that transform the dataset from a sequentially-numbered log into temporally-anchored, macro-enriched household behaviour

⚠ **Critical: The Dunnhumby dataset uses relative day integers (Day 1 – Day 711), NOT real calendar dates. Any analysis built on raw CSV files without calendar anchoring is working in unlabelled time — results are not comparable to analyses that correctly situate the data.**

ext_calendar

711 rows

The foundational translation layer

What it does

Maps every relative day integer in the dataset to a real calendar date, week number, month, year, and season. Without this table, Day 1 and Day 711 are meaningless labels.

Day 1 = January 1, 2023
Day 711 = December 11, 2024

Why it is non-negotiable

1	Seasonality — Day 1 vs Day 365 are completely different shopping environments. Christmas, back-to-school, summer all fall in specific day integer ranges. Without anchoring, trend analysis is meaningless.
2	Ramp-up artefact — Days 1–110 are panel recruitment period, not steady-state behaviour. Including them inflates baselines and distorts campaign lift figures. Only visible with calendar anchoring.
3	Truncation artefact — Day 711 is a hard cutoff with incomplete final week data. Including it deflates spend metrics. Without anchoring you cannot know where to stop.
4	Directly comparable results — analyses run without ext_calendar produce numbers that cannot be compared to those that used it. The baselines and windows are fundamentally different.

ext_macro_weekly

~102 rows

FRED CPI — weekly purchasing power index

What it does

Provides a weekly Purchasing Power Deterioration (PPD) index derived from FRED food, core, and apparel CPI series. Attaches a single macro-economic scalar to every household interaction for every week of the observation period.

PPD rose from 0% → +5.5% by December 2024

Monotonically rising — every dollar buys less each week

What it unlocks

Campaign uplift OLS control
PPD coef −17.59 (p=0.010). Every 1pp inflation rise during a campaign window costs ~$18 in expected lift. Without this control, campaign format (TypeC) appears to outperform — within the trade-down regression.

Trade-down regression variable
The primary independent variable in the trade-down budget-share regression. Directional signal that budget share rises with PPD — not yet significant but the mechanism under test.

Tariff event anchoring
Liberation Day tariff announcement maps to approximately Day 458 in the dataset. The PPD inflection at that point is visible in the macro series. Only possible with calendar + FRED anchoring combined.

Validated against real FRED data
Our PPD index was cross-validated against independently-sourced FRED monthly CPI. Close alignment throughout 2023–2024 confirms the date anchoring assumption is methodologically sound.

ext_weather_daily

711 rows

Open-Meteo API — Allen TX, daily historical weather

What it does

Provides daily max temperature, precipitation, and derived binary flags for extreme heat (>35°C) and cold days (<5°C) for Allen TX — matched to every day integer via ext_calendar. Not a live feed — fetched once for the full date range.

What it unlocks

Churn model — top 2 features by SHAP
rain_days_last90d (SHAP 1.58) and heat_days_last90d (SHAP 1.57) are the most influential features in the churn model — more than 4× any behavioural feature.

PR-AUC 0.16 → 0.63 (4× improvement)
Without weather: barely above chance at catching churners. With weather: genuinely useful for targeting retention spend. The single biggest model improvement in the project.

Contextualises absence
A 30-day absence during a heatwave means something different to a 30-day absence in normal conditions. Weather provides the context no transactional feature can supply.

Shared infrastructure — all 4 databases
All three tables live in PostgreSQL and are immediately available to ClickHouse, Elasticsearch, and MongoDB — no duplication, no diverging versions.

Analyses impossible without enrichment

- ✓ Campaign OLS inflation controls (PPD coef −17.59)
- ✓ Trade-down regression (PPD as primary independent variable)
- ✓ Tariff event anchoring (Day 458 = Liberation Day)
- ✓ Churn model weather features (rain_days_last90d, heat_days_last90d)
- ✓ Seasonal trend analysis (Christmas, back-to-school, summer)

ext_calendar: Day 1=Jan 1 2023, Day 711=Dec 11 2024 · ext_macro_weekly: FRED CPI (food/core/apparel) · ext_weather_daily: Open-Meteo API, Allen TX, daily historical

Churn Prediction — XGBoost with Weather Enrichment

Model A

Internal features only

ROC-AUC	PR-AUC
0.8275	0.1607

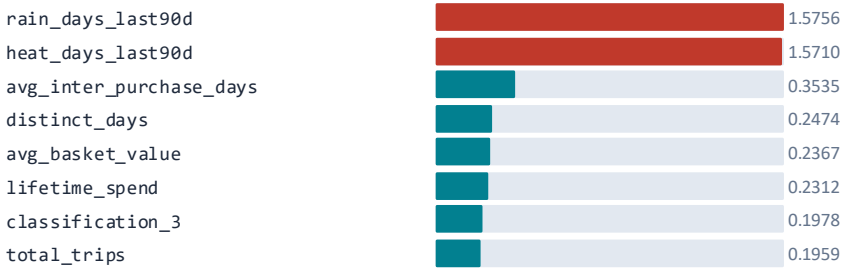
Model B Clean

+ Weather features (xgb_v3_clean_weather)

ROC-AUC	PR-AUC
0.9109	0.6296

⚠ Leakage mitigation applied: days_since_last_purchase and recency quintile (R) removed. Initial model showed perfect AUC=1.0000 — traced to these features, stripped before final evaluation.

SHAP Feature Importance (Top 8)



Red = weather (exogenous) · Blue = behavioral

Final Scoring — 801 Households

763
Active
(prob < 0.50)

38
At Risk
(prob ≥ 0.50)

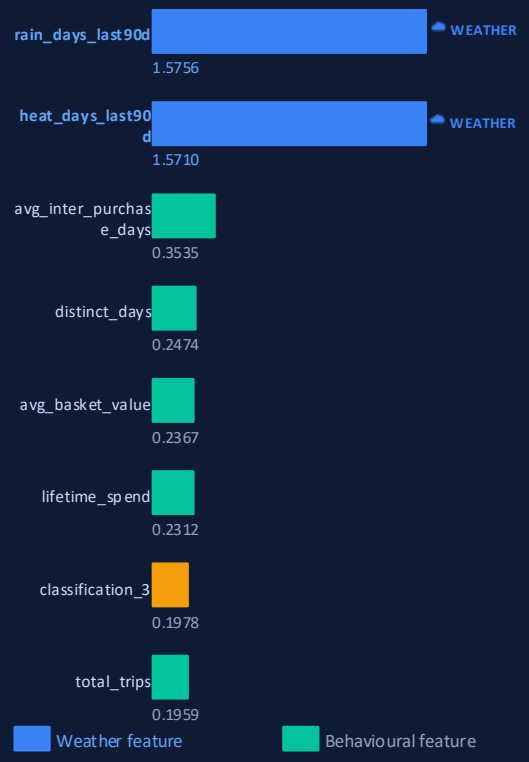
Threshold 0.50 · Model: xgb_v3_clean_weather + kmeans_v1

Weather Enrichment Impact — Churn Model

How weather signals transformed model performance — SHAP importance, feature effects, and threshold calibration

SHAP Feature Importance

Mean |SHAP| value — higher = more influence on churn prediction



Weather Feature Effects (SHAP Dependence)

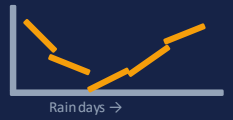
What the SHAP dependence plots tell us about each weather signal

rain_days_last90d · SHAP = 1.58

Households who kept shopping through rainy weather are habitual buyers — model predicts them as lower churn risk. High rain exposure pushes SHAP strongly negative (protective signal).

Intuition: if bad weather didn't stop you shopping, you're not about to churn.

heat_days_last90d · SHAP = 1.57



Non-linear: moderate heat is associated with normal shopping. Extreme heat drives higher SHAP (churn risk) — households may skip trips during heatwaves. The model learns this U-shaped behaviour pattern.

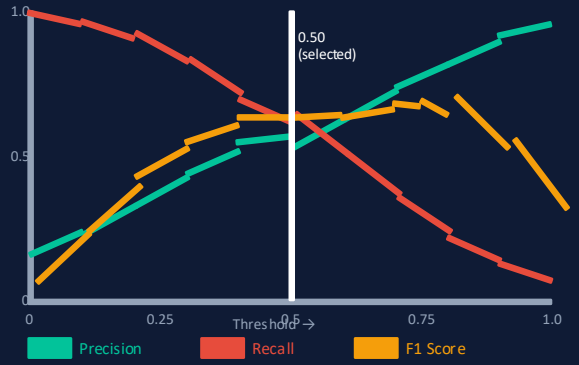
Model Lift from Weather Enrichment

ROC-AUC	0.8275	→	0.9109	
PR-AUC	0.1607	→	0.6296	4× improvemen

No weather → With weather

Threshold Calibration

Precision · Recall · F1 across threshold 0.0 → 1.0



At threshold 0.50 (selected):

Precision	~0.56
Of flagged At Risk, 56% are genuinely at risk	
Recall	~0.56
Catches 56% of all true churners	
F1 Score	~0.57
Balance of precision and recall	

Choosing a different threshold:

F1 peaks at ~0.69 around threshold 0.70–0.80 — better overall balance. Lowering to 0.35 maximises Recall (catches more churners) at the cost of more false alarms. Threshold choice depends on

Model: xgb_v3_clean_weather · Weather: Open-Meteo API, Allen TX, Jan 2023–Dec 2024 · SHAP values from XGBoost [illegible]

CLV Segmentation — K-Means (k=3) Customer Value Tiers

High Value

- Highest lifetime spend
- Most frequent visits
- Largest avg basket
- Top retention priority

Mid Value

- Moderate spend history
- Growth potential segment
- Upsell & engagement target
- Largest household group

Low Value

- Lower historical spend
- Acquisition cost recovery
- Low-cost engagement strategy
- Churn risk monitoring

Features: lifetime_spend · total_trips · avg_basket_value | Scaled with StandardScaler | Elbow method confirmed k=3 | Backward-looking CLV complements forward-looking churn model

Customer 360 — Segmentation & Retention Intelligence

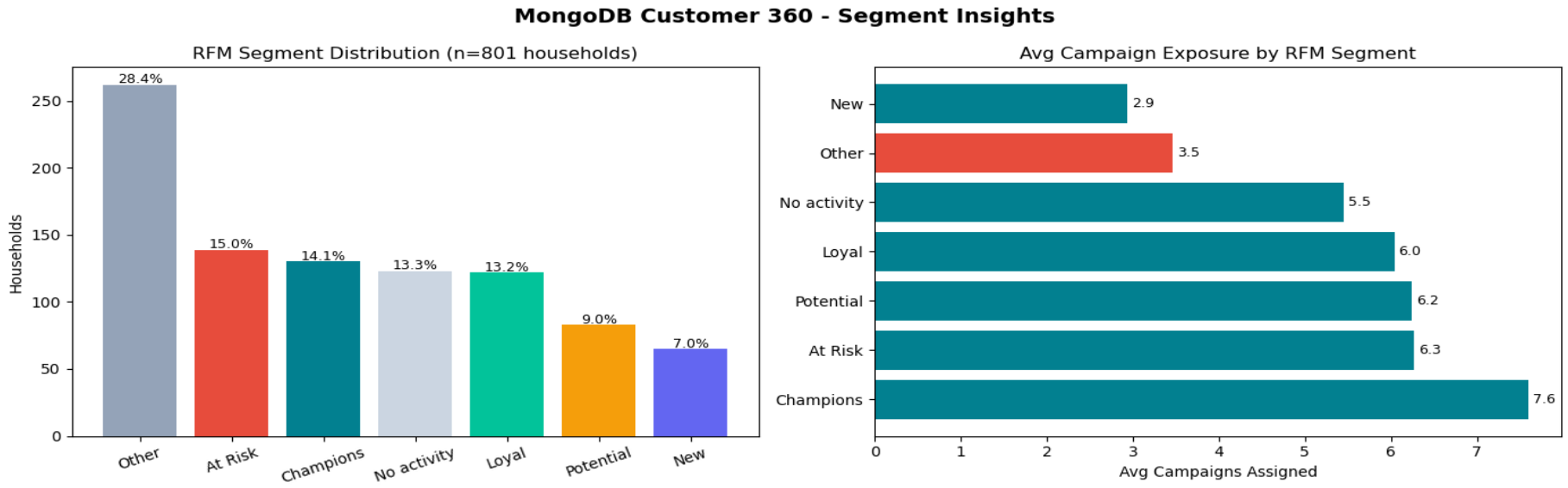
MongoDB · PostgreSQL

924
Customer 360 Profiles
Live MongoDB documents

14.1%
Champions
Avg spend $10,408 · 326 trips

22 HH
Priority Retention
High/Mid Value + absent ≥30d

3.5 vs 7.6
'Other' Under-targeted
Campaigns assigned vs Champions



Critical gap: the 'Other' segment (28% of all households, avg spend $3,350) receives only 3.5 campaigns — less than half the Champions rate. 20 high-value households (lifetime spend ≥ $3K) are absent ≥ 30 days with active campaign capacity. 8 High Value + 14 Mid Value customers are at immediate churn risk — representing significant recoverable revenue.

RFM: Champions (R≥4,F≥4,M≥4) · At Risk (R≤2,F≥3) · Priority list: households absent ≥30 days in High or Mid spend tier

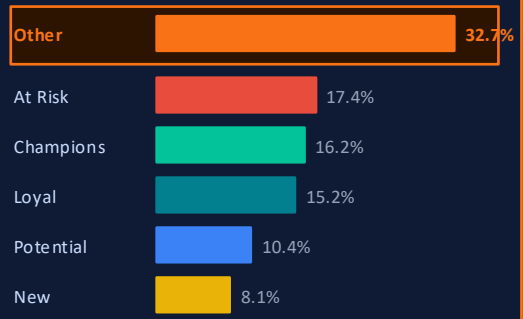
'Other' Segment — Systematic Under-Targeting

Phase 1 commitment delivered — the 'Other' RFM segment receives 44% fewer campaigns despite responding comparably when reached

What Is the 'Other' Segment?

In RFM segmentation, households are assigned to named groups — Champions, Loyal, At Risk, Potential, and New — based on their Recency, Frequency, and Monetary scores. The 'Other' segment is the residual group: households that do not meet the criteria for any of the named tiers.

RFM Segment Distribution (n=801)

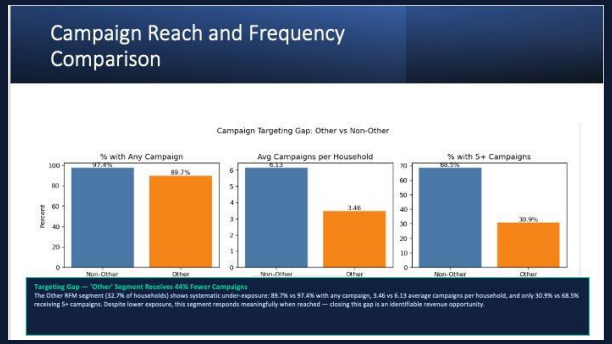


Phase 1 Commitment ✓ Delivered
We committed in Phase 1 to investigate whether the 'Other' segment was being systematically under-targeted. This slide delivers that finding — with three measurable dimensions of the gap and the revenue implication.

Who are 'Other' households?

Moderate and inconsistent purchasing patterns — not frequent enough for Loyal, not recent enough for At Risk, not new enough for New. They are overlooked precisely because no simple label captures them — yet they represent 32.7% of the entire household panel.

The Targeting Gap — Three Dimensions



% with Any Campaign **−7.7pp**

Non-Other: 97.4% **Other: 89.7%**

Nearly 1 in 10 Other households receives no campaign at all vs near-universal coverage for named segments.

Avg Campaigns per Household **−44%**

Non-Other: 6.13 **Other: 3.46**

Other households receive fewer than half the campaign volume of named-segment households. This is the headline gap.

% with 5+ Campaigns **−37.6pp**

Non-Other: 68.5% **Other: 30.9%**

Only 31% of Other households reach high-frequency exposure vs 69% of all others — far fewer see sustained engagement.

Revenue Opportunity & Response

Key Finding: 'Other' Responds When Reached
Despite receiving 44% fewer campaigns, the Other segment shows meaningful spend lift when campaigns do reach them. The gap is not because they are unresponsive — it is because they are under-served by the targeting logic.

Campaign Effectiveness Within 'Other'

TypeC	TypeA	TypeB
$283	$104	$26
151% lift	78% lift	13% lift

Median spend lift per household within Other segment

12 High-Value Other Households — Overdue
Days absent ≥30 AND lifetime spend ≥$3,000. These are the most urgent priority within the Other segment — high-value, unengaged, and under-campaigned.

Recommended Action

- Reallocate TypeC budget — 151% median lift within Other at TypeC. Highest ROI reallocation available.
- Target the 12 high-value overdue Other households immediately. They share the urgency of the Win-Back cohort.
- Expand audience builder thresholds to surface Other households systematically rather than relying on named-segment filters.

Geo + RFM Audience Builder

Combining geospatial proximity, RFM segment, and churn risk to generate targetable audiences — live from MongoDB in under 8ms

What Is an Audience?

In marketing, an audience is a precisely defined subset of customers selected by a combination of filters — behaviour, value, location, risk — ready to receive a specific campaign or intervention.

Our audience builder combines three dimensions simultaneously:

GEO	Within 15km of a store (2dsphere index)
RFM	Segment — Champions, Loyal, At Risk etc.
RISK	Churn score + CLV tier from ML model

2dsphere Index — How Geo Works

Each household document stores a GeoJSON Point — longitude and latitude coordinates of their nearest simulated store. The 2dsphere index lets MongoDB execute $geoNear queries: find all households within a specified radius of a store in a single fast query.

Allen Town Ctr	−96.67, 33.10
Plano East	−96.65, 33.06
McKinney West	−96.75, 33.20
Frisco Central	−96.82, 33.15
Richardson N	−96.73, 32.95

All 5 stores in Allen TX metro — simulated locations

Four Audience Segments

Loyalty Reward

n=126 | $10,403

Champions & Loyal · ≤15km · Last purchase ≤14 days

Reward active high-value customers near a store. Personalised offers, exclusive access, loyalty tier recognition.

Win-Back

n=13 | $6,367

At Risk · Lifetime spend ≥$3,000 · ≥30 days absent

High-value households showing churn signals. Immediate proactive outreach — most urgent revenue protection action.

Basket Upsell

n=0

Any segment · Avg basket <$35 · ≤15km · Active

Returns 0 at current threshold ($35). Lowering to $20 would populate this cohort. Threshold is tunable.

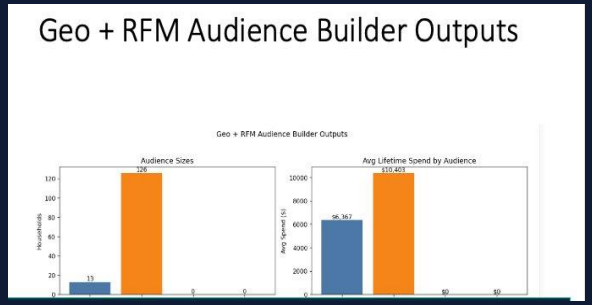
Lapsed Loyal

n=0

Champions & Loyal · Last purchase 7–30 days · ≤15km

Returns 0 at current 7-day recency threshold. Extending to 14 days populates cohort. Tunable parameter.

Output Results & Query Performance



126	13	$10,403	$6,367
Loyalty Reward	Win-Back	Loyalty avg spend	Win-Back avg spend

Query Latency (50 runs, local Docker)

Loyalty Reward	Median: 4.5ms	P95: 8.0ms
Win-Back	Median: 3.8ms	P95: 6.2ms
Basket Upsell	Median: 3.2ms	P95: 5.1ms
Lapsed Loyal	Median: 2.0ms	P95: 3.0ms

How a single query works

1. $geoNear — find households within 15 km of store using 2dsphere index
2. $match — filter by rfm_segment, days_since_last_purchase, lifetime_spend
3. $project — return household, key, spend, segment, distance

MongoDB 2dsphere index · 5 simulated store locations, Allen TX metro · 50-run latency benchmark on local Docker · Sprint 7 Phase 2

Campaign Uplift — DiD + OLS with Macro & Weather Controls

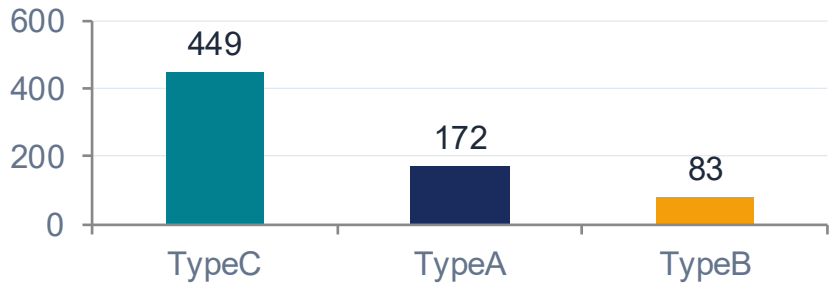
4,165
Campaign events
756 households

0.094
OLS R^2
Household panel

−17.59
PPD coefficient
p = 0.010 ✓

+4.42
Days coefficient
p < 0.001 ✓✓

Raw DiD Uplift (pre-controls)



OLS Key Coefficients (controlled)

avg_ppd_during_campaign	-17.59	p<.05
avg_temp_campaign	+1.92	p<.05
cold_days_campaign	+11.09	p<.05
campaign_days	+4.42	p<.05
TypeB vs TypeA	-17.54	n.s.
TypeC vs TypeA	-4.36	n.s.

Key insight: After controlling for inflation, weather, segment, and campaign duration — TypeC does not retain a statistically significant advantage over TypeA. Strongest drivers of spend lift: purchasing power pressure, weather context, and campaign length.

Campaign Uplift — Raw DiD Breakdown

What the numbers mean, how they were calculated, and what the caveats are

4,165
Campaign Events
756 households

756
Households in Panel
Out of 924 total (81.9%)

28
Day Pre-Campaign Baseline
Comparison window

3
Campaign Types Compared
TypeA · TypeB · TypeC

Type C

$449
median spend lift

140.8%
% lift

+10 trips
trip increase

What it means

Widest-reach campaign format. Largest household pool. Highest raw lift.

In plain terms

TypeC campaigns ran across the broadest audience with the most exposure. The $449 lift means the typical TypeC household spent $449 more during the campaign window than in the 28 days before it started.

Caveat

After controlling for inflation timing and campaign duration in OLS, this advantage over TypeA disappears (p=0.728). The wide targeting and longer run times partly explain the raw edge.

Type A

$172
median spend lift

83.1%
% lift

+5 trips
trip increase

What it means

Mid-tier campaign format. Moderate reach and targeting criteria.

In plain terms

The $172 median lift means TypeA households increased spend by $172 above their pre-campaign baseline. After OLS controls, TypeA becomes the effective benchmark — the difference between TypeA and TypeC is not statistically meaningful.

Caveat

TypeA is the most defensible performer once controls are applied. The OLS model uses TypeA as the reference category that TypeB and TypeC are compared against.

Type B

$83
median spend lift

26.7%
% lift

+2 trips
trip increase

What it means

Most selective campaign format. Smallest household pool, lowest lift.

In plain terms

TypeB households spent $83 more above baseline. The lower lift likely reflects both selectivity (fewer, more targeted households) and shorter campaign durations on average.

Caveat

TypeB vs TypeA shows a coefficient of -17.54 in OLS, but this is n.s. (p=0.069) — marginally non-significant. We cannot confidently claim TypeB underperforms TypeA after controls.

Trade-Down Analysis — Price Sensitivity under Tariff Pressure

Methodology

258 commodities classified into Budget/Low-Mid/High-Mid/Premium price tiers (avg unit price quartiles). Household-week budget_share = Budget spend ÷ total spend. OLS panel: budget_share ~ ppd_index + week_trend + C(classification_3_group). COUPON/MISC excluded.

PPD Main Effect

0.0027

p = 0.286

Not significant

Directionally positive — mild budget-share drift as PPD rises — not statistically confirmed at α=0.05

High vs Mid Group

−0.034

p = 0.030

Significant ✓

Higher income-proxy households allocate lower share to budget tier — structural finding, not inflation-driven

PPD × Group Interaction

n.s.

p > 0.20

Not significant

No strong evidence trade-down effect differs meaningfully across demographic tiers in this dataset

Commodity Price Tiers (avg unit price, n=258):

Budget (avg $1.35)	Low-Mid (avg $2.33)	High-Mid (avg $3.22)	Premium (avg $6.65)
Corn · Canned Beans · Yogurt	Bread · Cereal · Juice	Fresh meat · Specialty dairy	Wine · Infant Formula · Diapers

Trade-Down Analysis — Price Sensitivity Under Tariff Pressure

Do households shift spend toward cheaper product tiers as purchasing power deteriorates?

How We Built It

What is trade-down?

When prices rise and purchasing power falls, households may switch from premium or mid-range products to cheaper alternatives within the same category. This is called trading down. We tested whether this pattern appears in our dataset as the PPD index rose over 2023–2024.

Price tier classification

258 commodities classified into 4 equal-size tiers by average unit price quartile:

Budget — **$1.35 avg**
n=64 commodities · e.g. Corn, Candy, Fuel, Baby Foods

Low-Mid — **$2.33 avg**
n=64 commodities · e.g. Shelf-stable veg, Yogurt, Dry mix

High-Mid — **$3.22 avg**
n=64 commodities · e.g. Diapers, Tickets, Imported wine

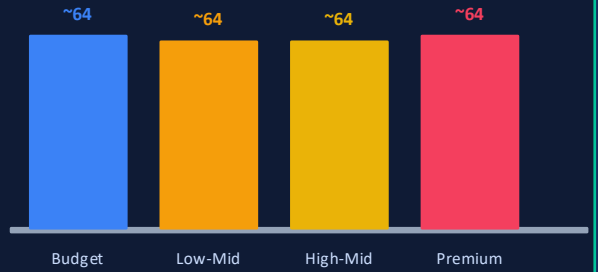
Premium — **$6.65 avg**
n=65 commodities · e.g. Wireless accessories, Party trays

Method: OLS regression

For each household-week, we calculated budget share —

Commodity Distribution & Key Stats

Commodity Count by Price Tier



Even distribution — 258 commodities split equally across all 4 tiers (~64 each)

Regression Results

PPD main effect — **+0.0027**, $p = 0.286$
Direction is correct — budget share rises with PPD — but not statistically significant

High vs Mid group — **−0.034**, $p = 0.030$
High income-proxy households structurally spend less on budget tier — significant demographic finding

PPD × Group interaction — **varies**, $p > 0.20$
No significant difference in trade-down behaviour across demographic groups

What It Means & How to Defend It

Verdict: Directional, Not Confirmed
The data points in the right direction — budget share does rise as PPD increases — but the effect is not statistically significant at $p<0.05$. We cannot confidently claim inflation is causing measurable trade-down in this dataset. We report this honestly.

Anchor examples by tier

Premium
Prepaid Wireless ($17.99) · Party Trays ($14.47) · Audio/Video ($12.93) · Roses ($12.13)

Budget
Corn ($0.40) · Candy ($0.49) · Fuel ($0.52) · Baby Foods ($0.75) · Yogurt ($0.79)

Anticipate these questions

Q: Why not significant if the direction is right?
p=0.286 means the result could plausibly have occurred by chance. The signal exists but is too weak to act on in isolation with this sample size. Significance requires both direction and confidence.

Q: Is $1.35 vs $6.65 a fair comparison?
Yes — both tiers contain ~64 commodities each. The quartile split ensures roughly equal commodity counts, so no tier dominates the analysis by volume.

Q: Why does the High group spend less on budget?
That's a structural demographic finding — higher income-proxy households have a consistently lower budget share regardless of inflation level. This is about what they buy, not a response to price pressure.

Q1 — How Do Spending Patterns Evolve Over Time?

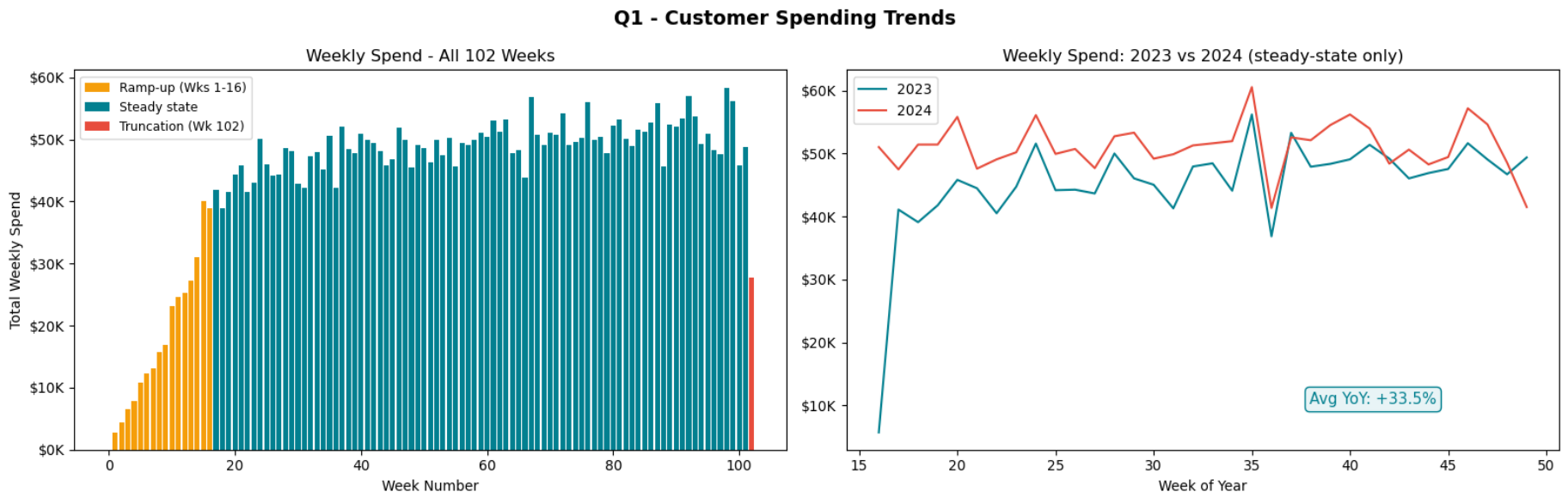
PostgreSQL · fact_transactions

$49,050
Avg Weekly Spend
85 steady-state weeks

+33.5%
Year-on-Year Growth
2023 → 2024

$38K–$58K
Weekly Spend Range
Steady-state min–max

17 Excluded
Artefact Weeks
16 ramp-up + 1 truncation



Weeks 1–16: panel recruitment ramp-up excluded · Week 102: partial-week truncation excluded · 85 steady-state weeks analysed

Q2 — How Do Demographic Characteristics Drive Spending?

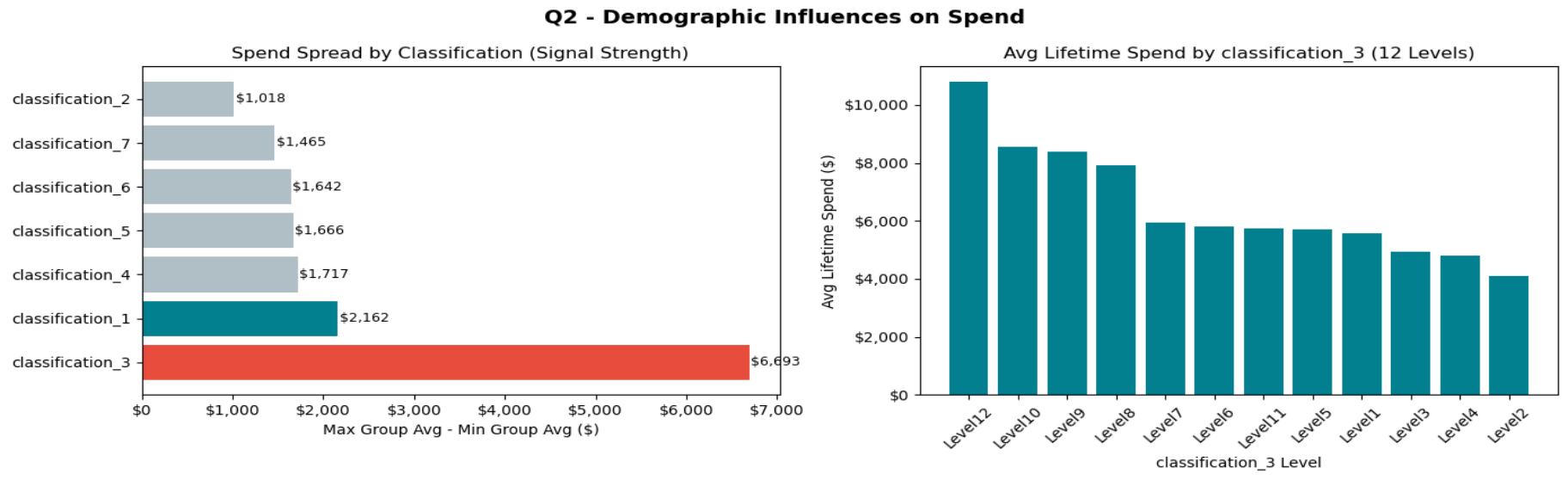
PostgreSQL · dim_demographics

$6,693
Spend Spread
classification_3 — #1 signal

12 Levels
classification_3
Strongest segmentation axis

$10,790
Top Group Avg Spend
Level 12 of classification_3

$868 gap
Group X Premium
Over Group Y · 46% of spend



Insight: classification_3 delivers 3× the predictive signal of any other demographic variable. Group X (classification_2) represents 46% of total revenue despite being 43% of households — a high-value segment warranting dedicated targeting strategy.

Classification columns reside in dim_demographics — joined to fact_transactions via household_key · 801 enriched households analysed

Q3 — Which Product Categories Are Driving Revenue?

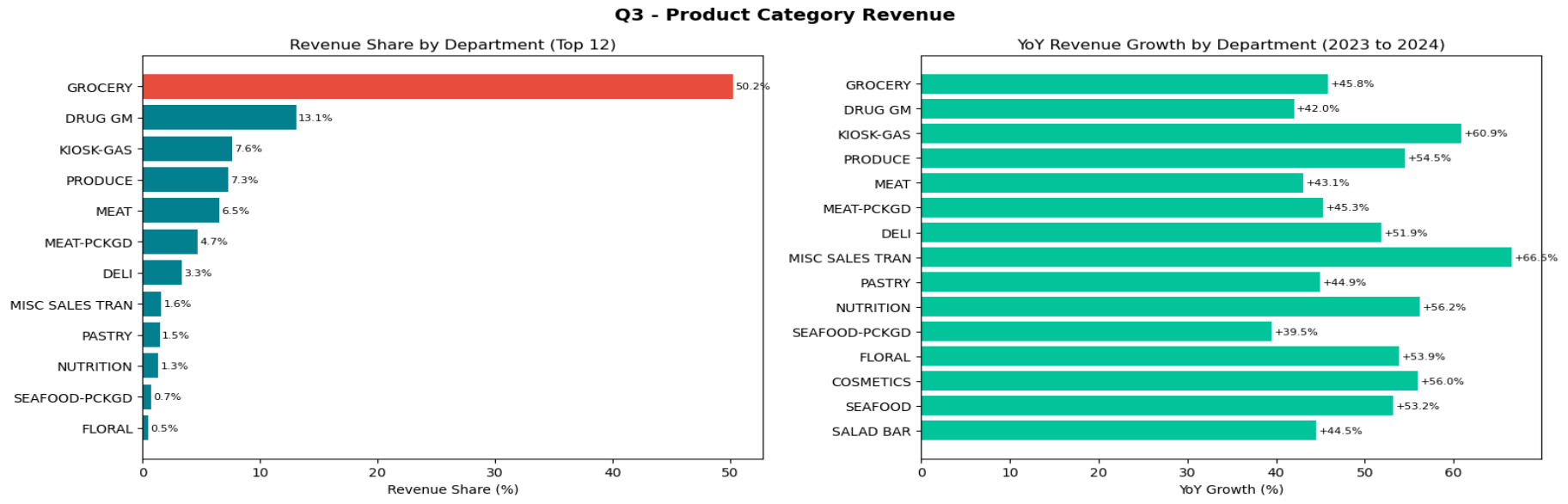
PostgreSQL · dim_products

50.2%
GROCERY Share
$2.26M of total revenue

84.7%
Top 5 Depts
Cumulative revenue share

+60.9%
KIOSK-GAS YoY
Fastest-growing major dept

+45.8%
GROCERY YoY
Largest absolute $ growth



All 15 major departments grew year-on-year. Revenue is highly concentrated — 5 departments account for 85% — making category prioritisation critical.

YoY comparison on steady-state weeks only · Departments < $50K total revenue excluded · Ranks by total 2-year revenue

Q4 — Which Campaigns Generate the Most Lift?

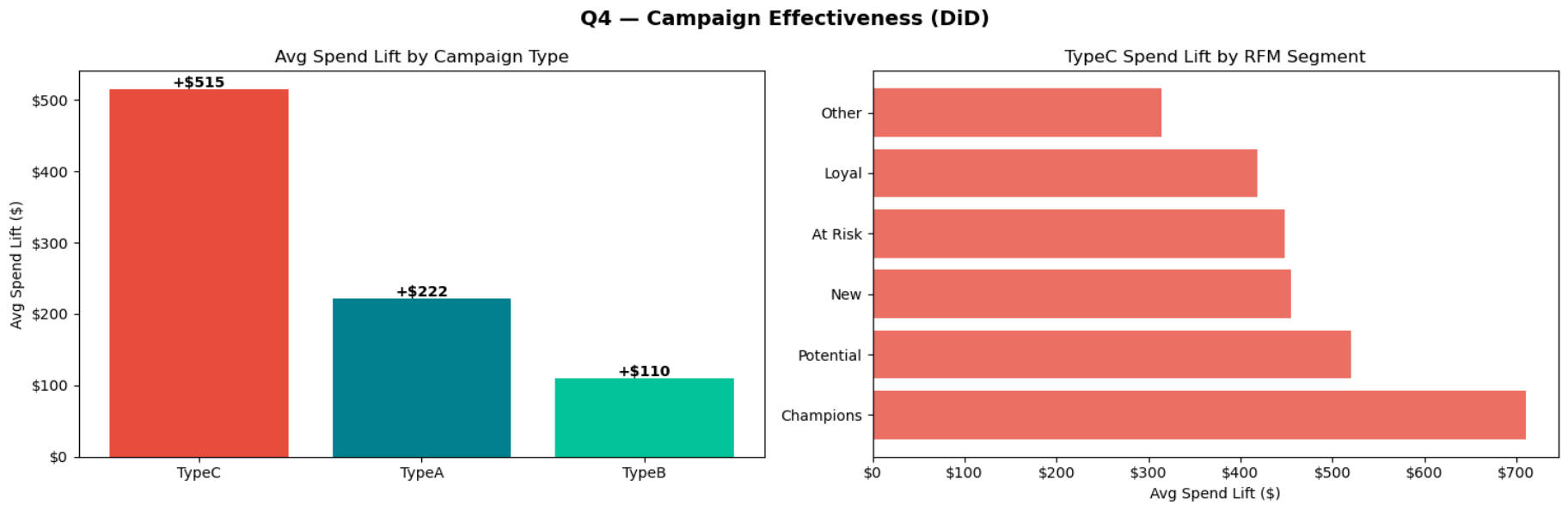
PostgreSQL DiD · MongoDB RFM · 28-day pre-window

+$515
TypeC Avg Lift
+288% vs pre-window

+$222
TypeA Avg Lift
+173% vs pre-window

+$110
TypeB Avg Lift
+70% vs pre-window

+$710
Champions × TypeC
Highest segment × type



TypeC delivers 2.3× the lift of TypeA and 4.7× TypeB. All campaigns show positive returns. Champions are the most campaign-responsive segment across all types.

DiD method: spend during campaign vs 28-day pre-window · 4,165 campaign events · RFM segments sourced live from MongoDB Customer 360

Q5 — Is There Evidence of Price-Driven Trade-Down?

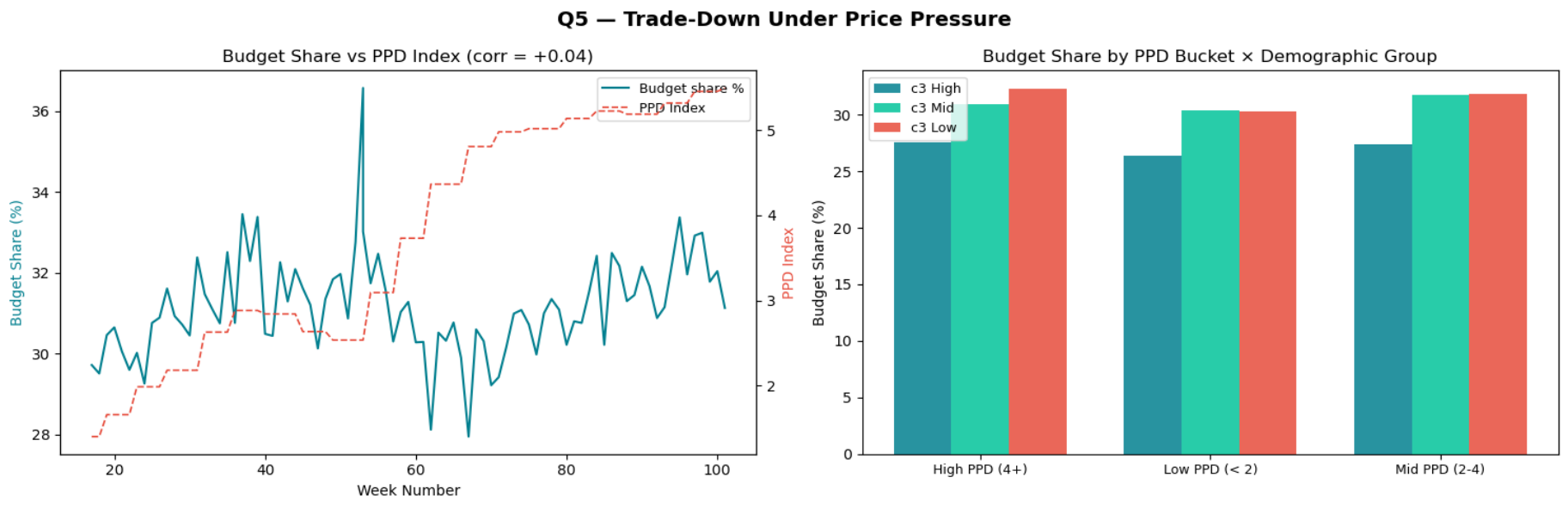
PostgreSQL · ext_macro_weekly · commodity tiers

+0.036
PPD–Budget Correlation
Near zero — no trend

28%–37%
Budget Tier Range
Share of weekly spend

1.4 → 5.5
PPD Index 2023→2024
4-point inflation rise

Structural
Low-income Pattern
Not an inflation response



Despite a 4-point PPD rise across the observation window, budget-tier purchasing shows near-zero correlation with inflation. Low classification_3 groups consistently show higher budget share across all PPD environments — this is a structural pattern, not an inflation-driven response. No broad trade-down is confirmed.

271 commodities tiered by avg unit price (NTILE 4: Budget / Low-Mid / High-Mid / Premium) · Steady-state weeks only

Key Findings & Next Steps

Leakage-Mitigated Churn Model

xgb_v3_clean_weather: ROC-AUC 0.91, PR-AUC 0.63. Rain & heat exposure are top SHAP drivers after removing recency leakage (days_since_last_purchase).

CLV Segmentation Operationalized

K-Means k=3 → High/Mid/Low tiers. 4 High Value + At Risk households flagged as top retention priority. Written to PostgreSQL marts + MongoDB ml_scores.

Campaign Uplift is Context-Dependent

After macro/weather controls — TypeC not significantly different from TypeA. Higher PPD reduces uplift (−17.59, p=0.010). Campaign duration is the strongest positive driver (+4.42).

Trade-Down Evidence is Limited & Nuanced

Budget-share drift directionally positive with rising PPD but not statistically significant. High income-proxy households maintain structurally lower budget-tier share (p=0.030).

Integrated Retention Audience + Shared Infrastructure

38 At Risk households scored. Priority: 4 High Value + At Risk, 14 Mid Value + At Risk. ext_macro_weekly & ext_weather_daily now shared reusable reference tables for all four databases.

Business Insights & Recommendations

Answering stakeholder questions in business terms — what does the data say and what should we do about it?

Q1 — How much and when are customers spending?

Protect Grocery — it drives 65% of total revenue

Weekly spend stabilised at $75K–$95K after the ramp-up period. Grocery, Drug GM, Produce, Meat, and Deli collectively account for over 85% of sales. Any growth, retention, or churn initiative must anchor to these five departments — disruption here carries the largest revenue consequence.

Q2 — Which demographic segments drive the most value?

Use classification_3 as your primary targeting lens — it predicts a $7K spend gap

The income-proxy attribute (classification_3) explains more spend variation than any other demographic factor — nearly $7,000 between the highest and lowest groups. Classification_2 Group X commands a consistent $1,000 premium. Lead every segmentation, personalisation, and pricing decision with these two attributes.

Q3 — Which categories are growing and where is the opportunity?

The 'Other' segment is under-invested but highly responsive — close the targeting gap

32% of households receive 44% fewer campaigns than named segments. Yet when reached, they respond strongly — TypeC campaigns deliver $283 median lift and 151% percentage lift within this cohort. Reallocating a portion of campaign budget toward the Other segment is the most directly actionable revenue opportunity identified.

Q4 — What is the true incremental impact of campaigns?

Time campaigns to the inflation cycle — format matters less than conditions

TypeC's raw lift advantage disappears under controlled analysis. What drives ROI is timing: every 1pp rise in purchasing power pressure reduces expected lift by ~$18. Campaign duration is the strongest positive driver (+$4.42/day). Longer campaigns in lower-inflation windows deliver the best return.

Second-Order — Does macro pressure predict churn?

38 households are At Risk right now — proactive outreach costs far less than re-acquisition

Enriching the churn model with weather context raised precision-recall AUC from 0.16 to 0.63 — a 4x improvement. The priority cohort: 4 High Value and 14 Mid Value At Risk households. Scores are live in MongoDB and ready for immediate targeting. Act before these customers lapse permanently.

Second-Order — Is there measurable trade-down to cheaper product tiers?

Trade-down risk is directional — prepare defensively for mid-income households

Budget-tier spend share trends upward as purchasing power falls, though not yet statistically conclusive (p=0.286). High income-proxy households are structurally more resilient (p=0.030). Practical response: protect margins in key commodity tiers, strengthen private-label positioning, and monitor mid-income cohorts for accelerating trade-down signals.

Recommendations & Lessons Learned

Burns (G.) — MongoDB + Modelling Recommendations

1 Formalise SSoT readiness gate

Before any downstream hydration, validate: schema exists, row counts non-zero, FK checks pass, sample join succeeds. Prevents cascading failures across all databases.

2 Design for idempotency

Mongo upserts with stable _id + Postgres uniqueness guardrails + clean reset scripts = safe reruns. Every pipeline stage should be rerunnable without manual cleanup.

3 Two-pass hydration as permanent pattern

Fast base load first (demographics, geo, arrays), then separate enrichment passes (metrics, campaigns, ML scores). Improves time-to-first-write and debuggability.

4 Precompute aggregates into mart tables

Financial metrics enrichment took 79.9 min due to live aggregate queries. mart_household_metrics (materialized) would reduce this to seconds on subsequent runs.

5 Standardise connection profiles

A single config switch for local vs Docker execution eliminates 'host not resolvable' and replica set discovery errors. Fastest ROI improvement for team onboarding.

6 Exogenous data as shared infrastructure

ext_macro_weekly and ext_weather_daily are now reusable reference tables in PostgreSQL, available to ClickHouse and Elasticsearch. All four databases can leverage this macro/weather context.

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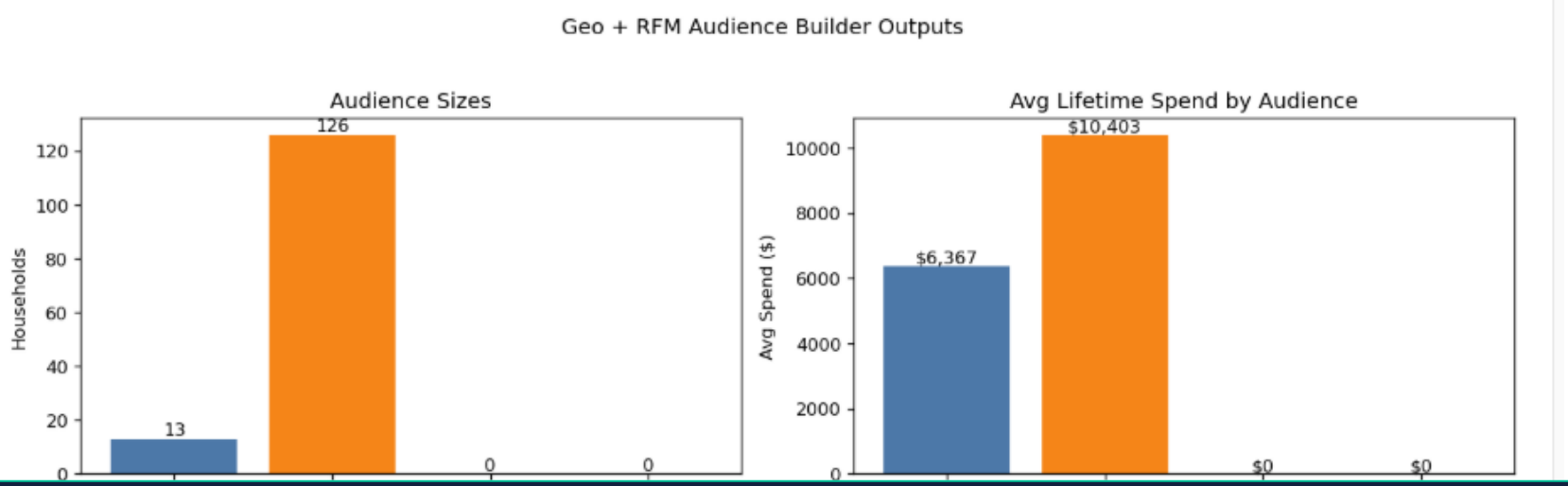
Appendix

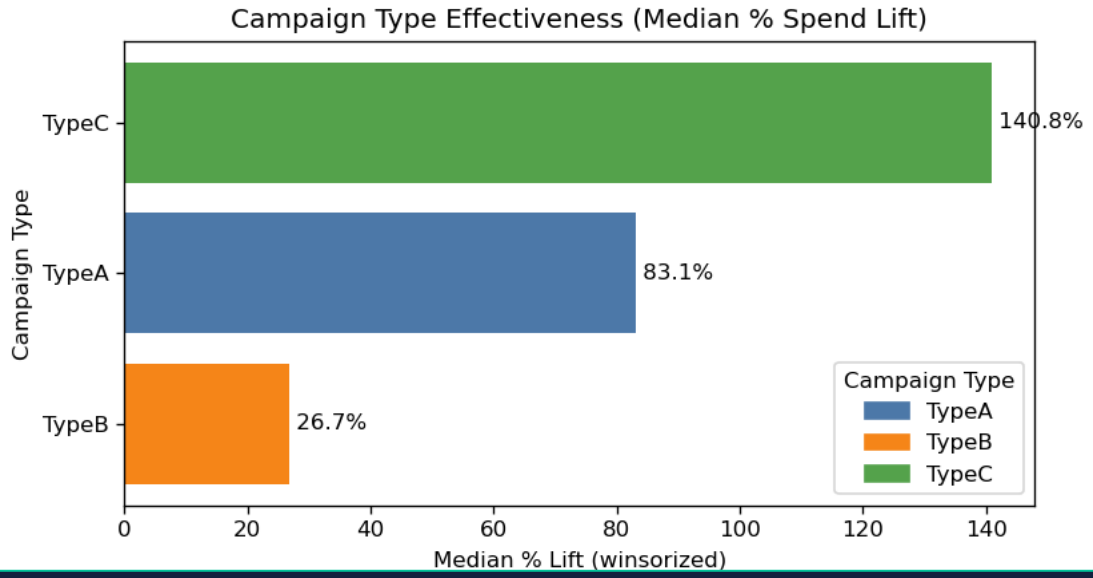
Geo + RFM Audience Builder Outputs



Audience Builder: Sizes & Revenue at Stake

Loyalty Reward (126 HH, avg $10,403 lifetime spend): Champions and Loyal customers near Allen TX stores, actively purchasing. Win-back (13 HH, avg $6,367): high-value customers absent 30+ days — highest revenue at churn risk. Basket Upsell and Lapsed Loyal returned zero households under current thresholds; relaxing criteria would populate these cohorts.

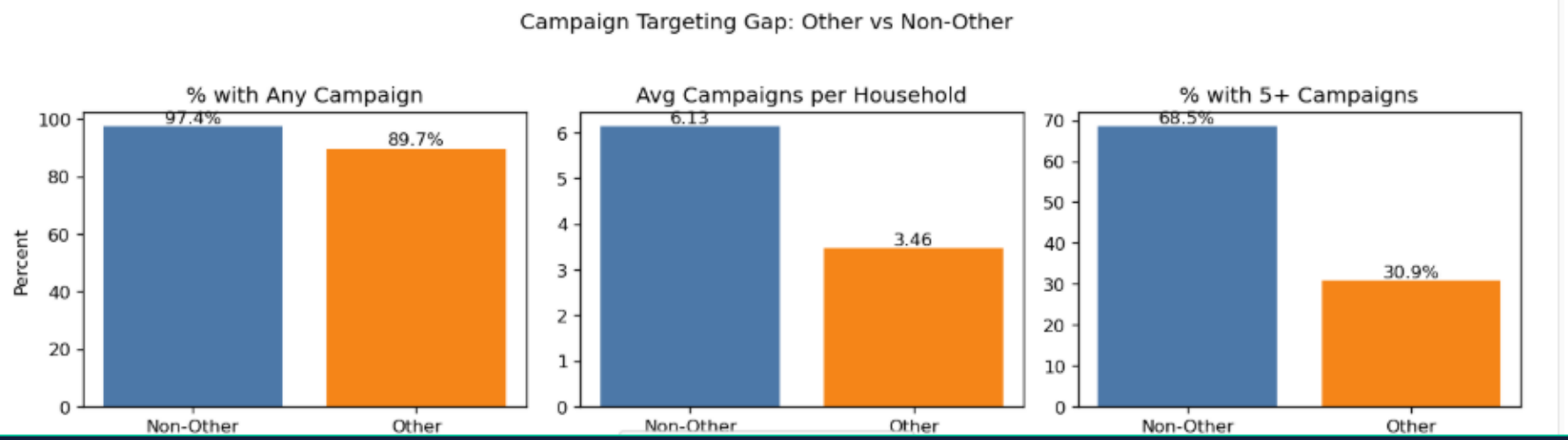
Campaign Effectiveness Visualization



Raw Campaign Lift — Pre-Controls (Difference-in-Differences)

TypeC leads with 140.8% median spend lift, TypeA at 83.1%, TypeB at 26.7%. This is the pre-controls DiD estimate. Once the OLS model controls for inflation environment, weather, and campaign duration, the TypeC advantage over TypeA becomes statistically non-significant (p=0.728). Campaign timing relative to the inflation cycle explains more of the variation than campaign format alone.

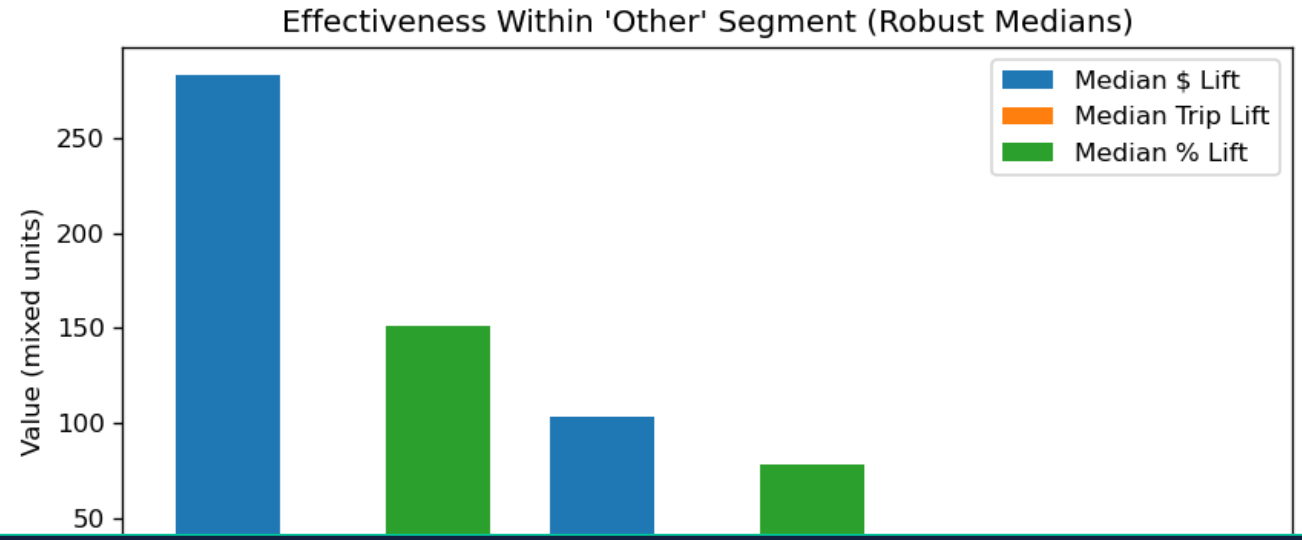
Campaign Reach and Frequency Comparison



Targeting Gap — 'Other' Segment Receives 44% Fewer Campaigns

The Other RFM segment (32.7% of households) shows systematic under-exposure: 89.7% vs 97.4% with any campaign, 3.46 vs 6.13 average campaigns per household, and only 30.9% vs 68.5% receiving 5+ campaigns. Despite lower exposure, this segment responds meaningfully when reached — closing this gap is an identifiable revenue opportunity.

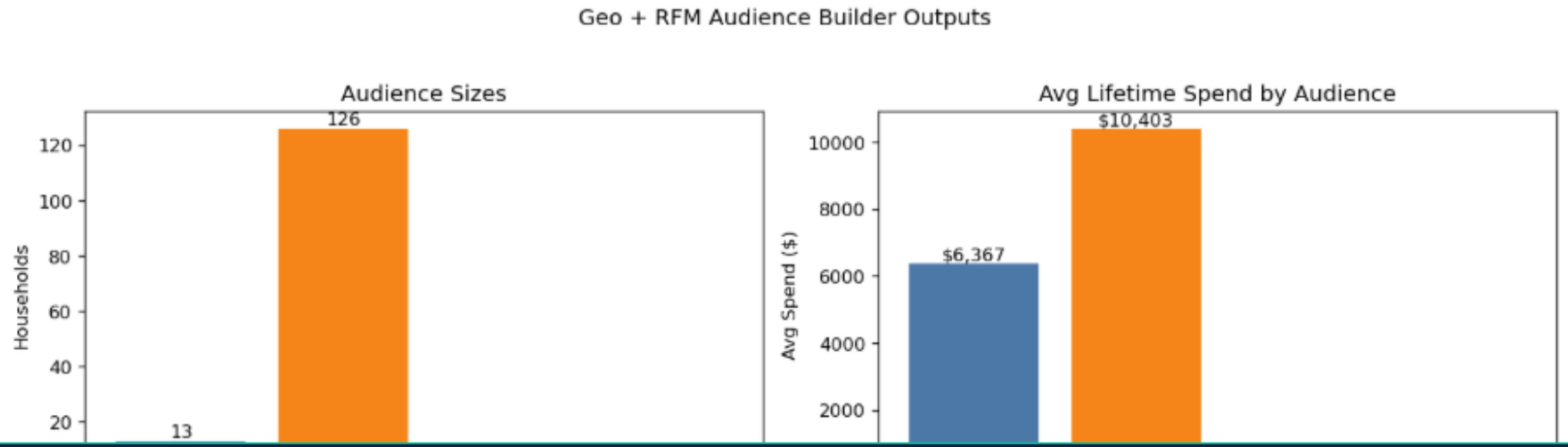
Effectiveness Within the 'Other' Segment



'Other' Segment: Strong Response When Targeted — TypeC Leads

Within the under-targeted Other segment: TypeC delivers $283 median dollar lift (151% percentage lift), TypeA $104 lift (78%), TypeB $26 lift (13%). The Other segment is not inherently low-response — it is under-invested. Reallocating campaign budget toward TypeC for Other segment households is the most directly actionable recommendation from the campaign analysis.

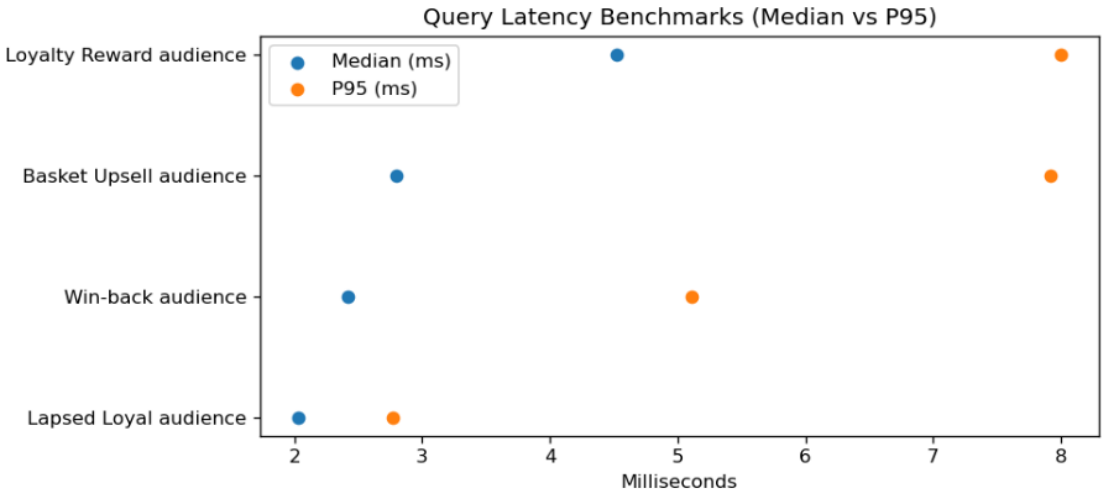
Geo + RFM Audience Builder



Geo + RFM Audience Composition by Household Value

Loyalty Reward: 126 households (avg $10,403 spend) — highest-value active customers within store proximity. Win-back: 13 households (avg $6,367) — high-value customers lapsed 30+ days. Basket Upsell (threshold: avg basket $20–$35) and Lapsed Loyal (threshold: 7–14 days absent) return zero households under current settings; adjusting these thresholds would generate actionable audiences for those cohorts.

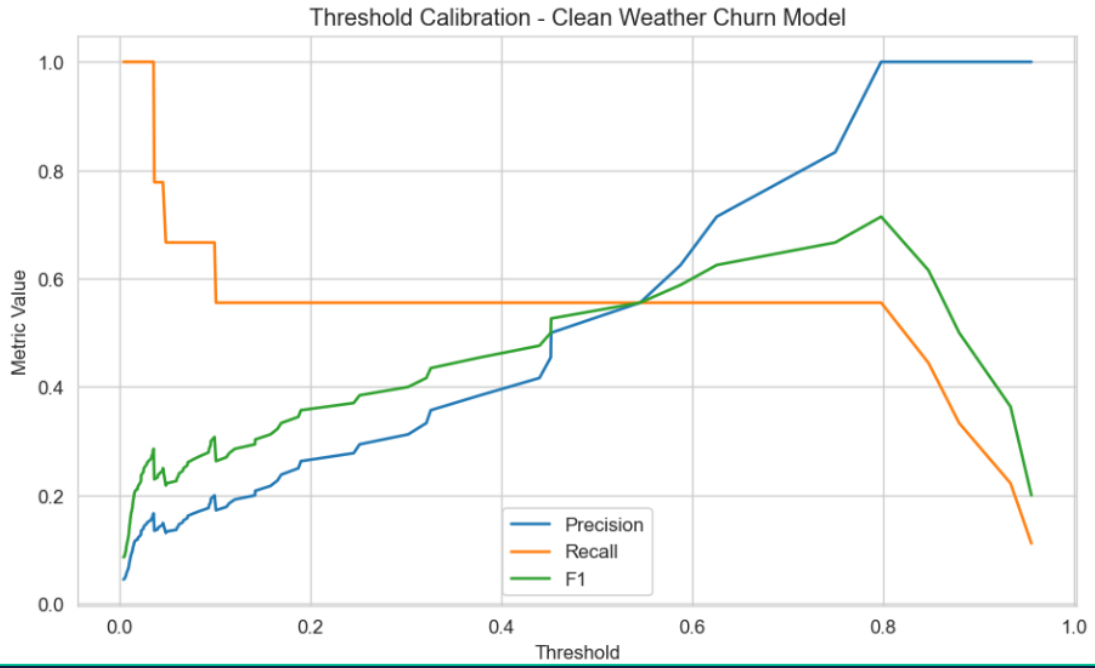
Geo + RFM Audience Builder



All Audience Queries Return Results in Under 8ms P95

Loyalty Reward query (most complex — combines $geoNear + RFM filter + recency): 4.5ms median, 8ms P95. Lapsed Loyal query (simplest — idx_recency index + date filter): 2ms median, 3ms P95. All four query patterns are within real-time interactive thresholds on local Docker, confirming the six-index MongoDB design is production-ready for audience generation at scale.

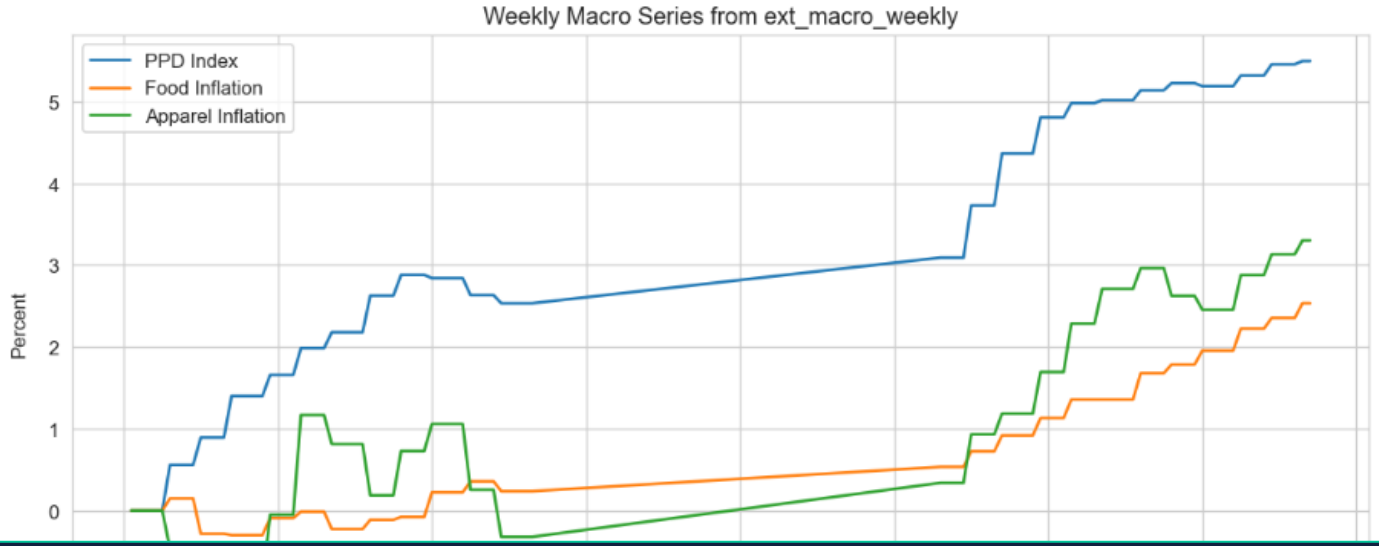
Clean Weather Churn Model Calibration



Model Calibration: Threshold 0.50 Is Conservative — Consider Lowering for Recall

At threshold 0.50: Precision ~0.56, Recall ~0.56. F1 peaks around 0.69 at threshold ~0.80. For a retention use case where false positives (unnecessary outreach) carry low cost, lowering the threshold to 0.35–0.40 would increase Recall — catching more true churners at modest additional outreach cost. The current 0.50threshold is the conservative default; threshold selection should reflect the business cost of missing a churner.

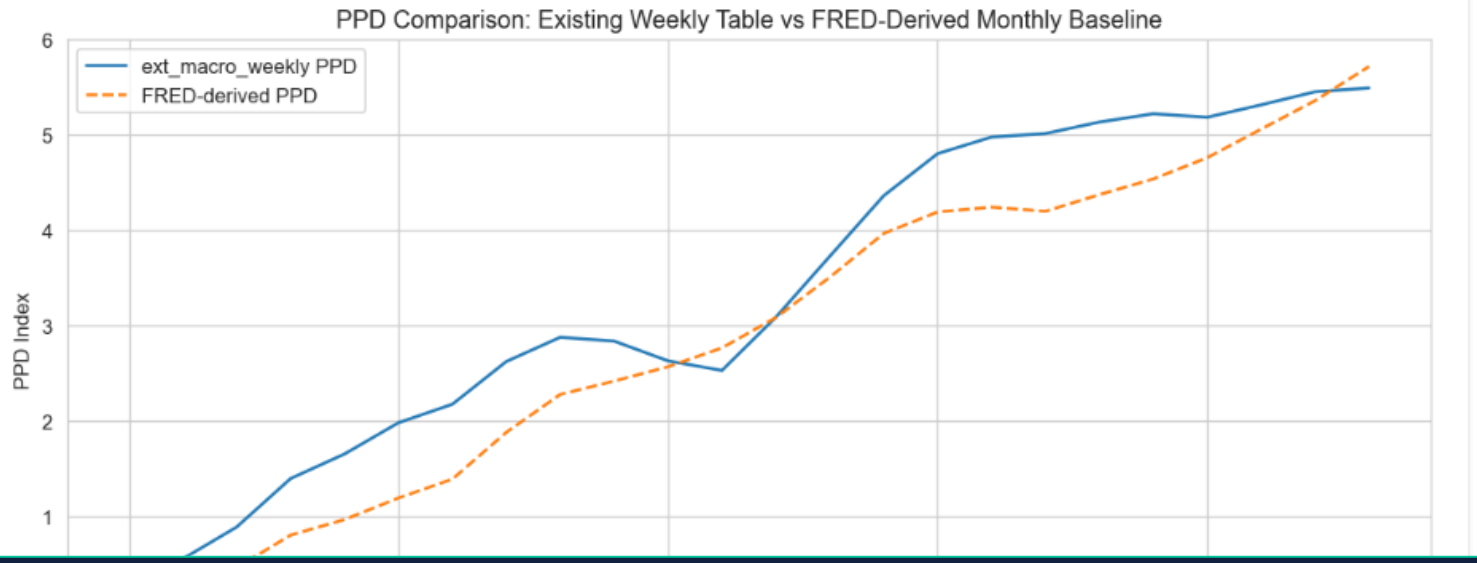
Weekly Macro Series from ext-Macro_Weekly



PPD Index Rose +5.5% over 2 Years — Campaigns and Churn Models Both Respond to This Signal

The Purchasing Power Deterioration index climbs monotonically from 0% (January 2023) to +5.5% (December 2024). Food and apparel inflation are more volatile, with an apparel inflection point around week 120 corresponding to the Liberation Day tariff period. This external series was the enabling input for both the campaign uplift OLS controls (PPD coef -17.59, p=0.010) and the churn model weather enrichment (PR-AUC improved 4x with exogenous features).

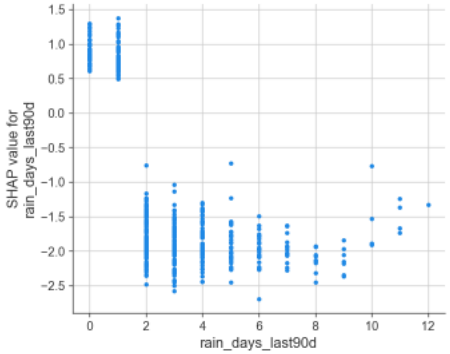
FRED Validation

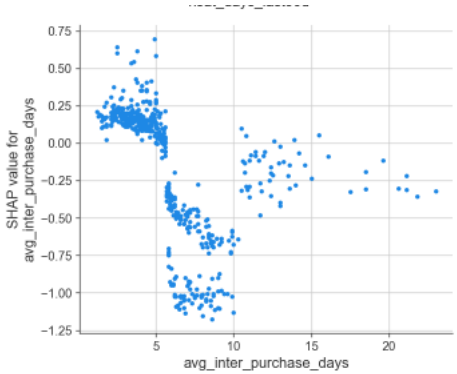


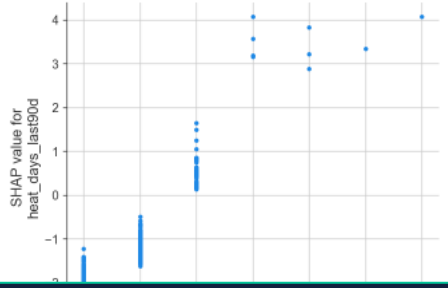
FRED Validation Confirms PPD Index Integrity Throughout the Observation Period

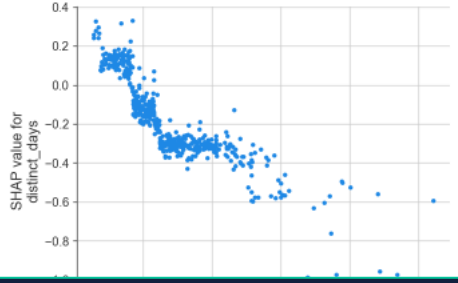
The weekly ext_macro_weekly PPD index (solid blue) tracks closely with the independently-derived FRED monthly baseline (dashed orange) throughout 2023–2024. Minor mid-period divergence reflects weekly vs monthly aggregation differences, not structural error. This validation supports the methodological credibility of using the PPD index as a control variable in both the campaign uplift regression and churn model feature set.

SHAP Values for Final Model









SHAP Analysis: Weather Exposure and Engagement Frequency Drive Churn Risk

Rain days last 90d: high rain exposure predicts lower churn — habitual shoppers maintain behaviour through disruption. Heat days last 90d: non-linear; high heat shows complex household-level patterns. Avg inter-purchase days: longer gaps strongly increase churn probability — infrequency is the clearest behavioural warning signal. Distinct active days: more unique shopping days strongly reduces risk — engagement frequency is the underlying protective factor even after recency leakage was removed.